INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

CLOTH, PLAIN WEAVE, NYLON: WATER REPELLENT TREATED AND UNTREATED

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE. This specification covers the requirements for four (4) Types, and fourteen (14) Classes of plain weave nylon cloth as specified.

1.2 Classification. This specification covers the following Types and Classes as specified.

1.2.1 Types.

Type I - Untreated – Air permeability requirement
Type II - Water repellent treated – Air permeability requirement
Type III - Water repellent treated – No air permeability requirement
Type IV - Water repellent/back coated

1.2.2 Classes.

Class 1 - Camouflage Green 483
Class 2 - Foliage Green 504
Class 3 - Tan 380
Class 4 - Tan 499
Class 5 - Coyote 476
Class 6 - Coyote 498
Class 7 - Snow White 506 (see.6.5)
Class 8 - Solid colors (with no spectral reflectance)
Class 9 - Universal Camouflage Pattern (UCP)

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

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1.2.2 Classes. - Continued

Class 10 - Operational Camouflage Pattern (OCP)
Class 11 - 3-Color Desert Camouflage
Class 12 - Marine Pattern (MARPAT) Woodland
Class 13 - Marine Pattern (MARPAT) Desert
Class 14 - Marine Pattern (MARPAT) Snow

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50199 - Thread, Polyester Core, Cotton or Polyester-Covered

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Copies are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY NATICK SOLDIER, RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

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2-1-2519 - Universal Camouflage Pattern – (UCP)
2-1-2592 - Operational Camouflage Pattern – (OCP)
2-1-2240 - Desert Camouflage Pattern (3 color)
2-1-2525 - Woodland MARPAT Pattern 4 color (Coyote 476)
2-1-2526 - Woodland MARPAT Pattern 4 color (Green 474 with EGA symbol)
2-1-2527 - Woodland MARPAT Pattern 4 color (Black 477)
2-1-2528 - Woodland MARPAT Pattern 4 color (Khaki 475)
2-1-2529 - Desert MARPAT Pattern 4 color (Light Tan 479)
2-1-2530 - Desert MARPAT Pattern 4 color (Urban Tan 478)
2-1-2531 - Desert MARPAT Pattern 4 color (Light Coyote 481 with EGA symbol)
2-1-2532 - Desert MARPAT Pattern 4 color (Highland 480)
2-1-2564 - Snow MARPAT Pattern

(Copies of drawings are available from the U.S. Army Natick Soldier Research Development and Engineering Center, ATTN: RDNS-SES-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

DOCUMENTS AND PUBLICATIONS

CODE OF FEDERAL REGULATIONS

27 CFR Part 21 – Formula for Denatured alcohol
40 CFR Parts (150-180) – Protection of Environment

(Copies are available online at http://www.access.gpo.gov.)

ENVIRONMENTAL PROTECTION AGENCY

EPA Pollutants/toxics subtopics

(Copies are available online at https://www.epa.gov/air-research/research-health-and-environmental

Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)
(40 CFR Part 162) State Registration of Pesticide Products

(Copies are available online at http://www.epa.gov/pesticides.)

FEDERAL TRADE COMMISION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available online at http://www.ftc.gov.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1 - Gray Scale for Color Change
AATCC Evaluation Procedure 2 - Gray Scale for Staining
AATCC Evaluation Procedure 8 - AATCC 9-Step Chromatic Transference Scale
AATCC Evaluation Procedure 9 - Visual Assessment of Color Difference of Textiles
AATCC Test Method 8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC Test Method 16.3 - Colorfastness to Light: Xenon
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 22 - Water Repellency: Spray Test
AATCC Test Method 61 - Colorfastness to Laundering, Accelerated
AATCC Test Method 81 - pH of the Water-Extract from Wet Processed
AATCC Test Method 118 - Oil Repellency: Hydrocarbon Resistance Test
AATCC Test Method 127 - Water Resistance: Hydrostatic Pressure Test
AATCC Test Method 135 - Dimensional Changes of fabrics after Home Laundering
AATCC Test Method 169 - Weather Resistance of Textiles Xenon Lamp Exposure

(Copies are available online at http://www.aatcc.org.)

ASTM INTERNATIONAL

ASTM D276 - Standard Test Method for Identification of Fibers in Textiles
ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D751 - Standard Test Methods for Coated Fabrics
ASTM D1683/D1683M - Standard Test Method Failure in Sewn Seams of Woven Fabrics
ASTM D3775 - Standard Test Method for Warp (End) and Filling (Pick) Count of Woven Fabrics
ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3886 - Standard Test Method for Abrasion Resistance of Textile Fabrics (Inflated Diaphragm Apparatus)
ASTM D3990 - Fabric Defects, Standard Terminology Relating to
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of documents are available online at http://www.astm.org.)

TAPPI T451 - Flexural Properties of Paper (Clark Stiffness)

(Copies are available online at http://www.tappi.org.)

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies are available online at http://www.asq.org.)

(Fabric Defect Replica Scales are available online at http://www.sdatlas.com) (see 6.6)

HEALTHCARE INFORMATION

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Inspections

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection The finished cloth shall be subjected to conformance inspection in accordance with 4.3.

3.2 Standard sample. The finished cloth shall match the standard sample for shade and appearance, and shall, unless otherwise indicated, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials.

3.4.1 Yarn. The yarn shall be a bright, high tenacity multifilament nylon.

3.5 Color. Unless otherwise specified on the drawing or procurement document, the color of the dyed and finished cloth shall be as stated in 1.2.2 or as specified in the contract (see 6.2). All colors shall match the standard sample for shade and appearance (see 6.1).

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3.5.1 Visual shade matching (all Classes). The color and appearance of the cloth specified in 3.5 shall match the standard sample when tested as specified in 4.5.3.

3.5.2 Colorfastness (all Classes). The finished cloth referenced in 3.5 shall conform to the colorfastness requirements listed in Table I when tested as specified in 4.5.3 and 4.6.2.

TABLE I. Colorfastness requirements (all Classes except Class 7).

Colors	Color Evaluation	Laundering (3 cycles) (min.)	Light (after 40 AFU or 170 kJ/(m^{2}nm)@420 nm) 1/, (min)	Crocking (wet and dry) (min.)	Weathering (after 40 AFU or 170 kJ/(m^{2}nm)@340 nm) 1/, (min)
Solid Shades	All colors	3-4	3-4	3.5	3-4
Universal Camouflage	All colors	3-4	3-4	3.5	---
Operational Camouflage Pattern	Dark Green 528, Brown 529, Dark Brown 561	3-4	3-4	3.5	---
	Tan 525, Olive 527, Dark Cream 559, Light Sage 560	3-4	3	3.5	---
3 Color Desert Camouflage	All colors	3-4	3-4	3.5	---
Woodland MARPAT	All colors except Black 477	3-4	3-4	3.5	---
	Black 477	3-4	3-4	1.5	---
Desert MARPAT	All colors	3-4	3-4	3.5	---
Snow MARPAT	All colors	3-4	3-4	3.5	---

1/ AATCC Fading Units (AFU)

3.5.3 Classes 1 through 8 (solid colors, except class 7). The Classes 1 through 8, except class 7 shall be dyed to the shade as stated in 1.2.2. All colors shall match the standard sample for shade and appearance, and shall, unless otherwise indicated, be equal to or better than the standard sample.

3.5.4 Class 9, Universal Camouflage print (UCP). The Class 9 finished cloth shall be either dyed to a ground shade matching Desert Sand 500 and then overprinted with the remaining

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two (2) colors for the Urban Gray 501 and Foliage Green 502 areas of the camouflage pattern or when the ground shade is not dyed to Desert Sand 500, then all three (3) colors of the camouflage pattern shall be printed to match all three (3) colors of the pattern.

3.5.5 Class 10, Operational Camouflage Pattern (OCP). The Class 10 finished cloth shall be either dyed to a ground shade matching Dark Cream 559 and then overprinted with the remaining six (6) colors for the Tan 525, Light Sage 560, Olive 527, Dark Green 528, Brown 529 and Bark Brown 561 areas of the camouflage pattern or when the ground shade is not dyed to Dark Cream 559 then all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern.

3.5.6 Class 11, Desert Camouflage Pattern (3-Color). The Class 11 finished cloth shall be either dyed to a ground shade matching Light Tan 492 and then overprinted with the remaining two (2) colors for the Light Brown 493 and Light Khaki 494 areas of the camouflage pattern or . when the ground shade is not dyed to match Light Tan 492, then all three (3) colors of the camouflage pattern shall be printed to match all three (3) colors of the pattern.

3.5.7 Class 12, Woodland Camouflage Pattern (MARPAT). The Class 12 finished cloth shall be either dyed to a ground shade matching Khaki 475 and then overprinted with remaining three (3) colors for the Green 474, Coyote 476 and Black 477 areas of the camouflage pattern or when the ground shade is not dyed to match Khaki 475, then all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern.

3.5.8 Class 13, Desert Camouflage Pattern (MARPAT). The Class13 finished cloth shall be either dyed to a ground shade matching Light Tan 479 and then overprinted with the remaining three (3) colors the Urban Tan 478, Highland 480 and Light Coyote 481 areas of the camouflage pattern or when the ground shade is not dyed to match Light Tan 479, then all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern.

3.5.9 Class 14, Snow camouflage Pattern (MARPAT) The Class 14 finished cloth shall be either dyed to a ground shade matching White Snow 506 and then overprinted with the remaining three (3) colors Light Snow Gray 507, Medium Snow Gray 508 and Dark Snow Gray 509 areas of the camouflage pattern or when the ground shade is not dyed to match White Snow 506 then all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern.

3.6 Pattern execution, Classes 9 through 14. The pattern on the printed finished cloth(s) shall match the standard sample with respect to design, colors, and registration of the respective areas. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid coverage; skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. When a standard sample is not available for pattern execution, a pattern drawing will be provided (see 6.4), and the pattern on the finished cloth shall match that of the drawing (see 2.2.2. and 6.4).

The pattern repeat for each style shall be as follows:

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Class 9 - Universal Camouflage - 36.00 (+1.25, -2.50) inches in the warp direction.
Class 10 - Operational Camouflage - 25.25 (+1.25, -2.50) inches in the warp direction and 68 inches maximum camouflage printed area in the filling direction
Class 11 - Desert Camouflage 3-Color) - 16.75 (+1.25, -2.25) inches in the warp direction
Class 12 - Woodland (MARPAT) - 27.25 (+1.25, -2.50) inches in the warp direction.
Class 13 - Desert (MARPAT) - 27.25 (+1.25, -2.50) inches in the warp direction.
Class 14 – Snow (MARPAT) - 36.00 (±1.50) inches in the warp direction and 60 inches (±1.50) in the filling direction.

3.7 Spectral reflectance, solid colors (except Class 8). The spectral reflectance values for solid shades, Class 1, through 7 shall conform to the requirements specified in Table II and Table IIA when tested as specified in 4.5.3. Solid colors for Class 8 when specified shall not include spectral reflectance requirements (see 6.2 for color).

TABLE II. Spectral reflectance (percent), solid shades Classes 1, 2, 3 and 4.

Wavelength, nanometers (nm)	Class 1 Camouflage Green 483		Class 2 Foliage Green 504		Class 3 Tan 380		Class 4 Tan 499	
	Min	Max	Min	Max	N/A	N/A	Min	Max
600	3	10	8	26	N/A	N/A	8	26
620	3	10	8	26	N/A	N/A	8	26
640	3	10	8	28	N/A	N/A	8	30
660	3	11	10	30	N/A	N/A	8	34
680	3	13	10	34	N/A	N/A	12	38
700	4	28	12	38	25	53	12	40
720	5	40	16	42	25	54	16	46
740	7	52	16	46	25	55	22	50
760	11	60	18	48	26	56	30	50
780	17	64	18	48	27	57	34	54
800	24	67	20	50	28	58	36	56
820	32	70	22	54	30	59	38	58
840	37	71	24	54	33	62	38	58
860	40	73	26	56	36	65	40	60

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TABLE II.A. Spectral reflectance (percent), solid shades Classes 5, 6 and 7.

Wavelength, nanometers (nm)	Class 5 - Coyote 476		Class 6 -Coyote 498		Class 7 - White Snow 506	
	Min	Max	Min	Max	Min	Max
600	8	20	80	80	80	80
620	8	20	80	80	80	80
640	8	20	78	80	78	80
660	8	20	78	80	78	80
680	10	30	78	80	78	80
700	18	50	78	80	78	80
720	22	54	78	80	78	80
740	30	56	78	80	78	80
760	35	58	78	80	78	80
780	40	62	78	80	78	80
800	55	80	78	80	78	80
820	55	80	78	80	78	80
840	55	84	76	99	76	99
860	60	84	76	99	76	99

3.7.1 Spectral reflectance for camouflage patterns (Classes 9-14). The spectral reflectance values for camouflage patterns shall conform to the requirements specified in Table IIB through Table IIG when tested as specified in 4.5.3.

TABLE II.B. Spectral reflectance (percent), Class 9.

Universal Camouflage Pattern (UCP)						
Wavelength, nanometers (nm)	Desert Sand 500		Urban Gray 501		Foliage Green 502	
	Min	Max	Min	Max	Min	Max
600	28	40	12	26	8	18
620	30	42	14	26	8	18
640	34	48	14	28	8	20
660	38	56	14	30	10	26
680	44	60	18	34	10	26
700	46	66	24	38	12	28
720	48	68	26	42	16	30
740	48	72	30	46	16	30
760	50	74	32	48	18	32
780	54	76	34	48	18	34
800	54	76	34	50	20	36
820	54	76	36	54	22	38
840	55	78	38	54	24	40
860	56	78	40	56	26	42

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TABLE II.C. Spectral reflectance (percent), Class 10.

Operational Camouflage Pattern (OCP)						
Wavelength, nanometers (nm)	Dark Cream 559 & Tan 525		Light Sage 560, Olive 527 & Brown 529		Dark Green 528 & Bark Brown 561	
	Min	Max	Min	Max	Min	Max
600	22	44	12	30	3	12
620	24	45	12	30	3	12
640	24	45	12	32	4	12
660	25	45	12	32	4	13
680	28	45	14	34	4	18
700	28	48	14	36	6	25
720	30	52	16	39	6	27
740	32	55	18	41	10	29
760	36	56	20	43	14	33
780	38	57	22	45	18	36
800	40	57	22	45	20	37
820	44	58	24	46	20	38
840	46	59	26	47	21	39
860	48	60	28	48	21	40

TABLE II.D. Spectral reflectance (percent), Class 11

3-Color Desert Camouflage Pattern						
Wavelength, nanometers (nm)	Light Tan 492		Light Brown 493		Light Khaki 494	
	Min	Max	Min	Max	Min	Max
700	38	53	19	36	25	48
720	38	58	20	36	25	52
740	39	62	20	36	25	54
760	40	66	21	36	26	56
780	41	72	21	38	27	57
800	43	76	22	43	28	58
820	45	76	23	45	30	58
840	48	78	24	46	33	58
860	50	78	25	46	36	59

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TABLE II.E. Spectral reflectance (percent), Class 12.

Marine Pattern (MARPAT) Woodland Camouflage Pattern						
Wavelength, nanometers (nm)	Coyote 476 and Khaki 475		Green 474		Black 477	
	Min	Max	Min	Max	Min	Max
600	8	20	3	9	N/A	N/A
620	8	20	3	9	N/A	N/A
640	8	20	3	9	N/A	N/A
660	8	20	3	12	N/A	N/A
680	10	30	3	16	N/A	N/A
700	18	50	5	32	N/A	20
720	22	54	7	44	N/A	30
740	30	56	12	52	N/A	33
760	35	58	18	56	N/A	33
780	40	62	26	56	N/A	34
800	55	80	34	56	N/A	34
820	55	80	42	60	N/A	35
840	55	84	44	60	N/A	35
860	60	84	44	60	N/A	35

TABLE II.F. Spectral reflectance (percent), Class 13.

Marine Pattern (MARPAT) Desert Camouflage Pattern						
Wavelength, nanometers (nm)	Light Tan 479		Light Coyote 481 and Highland 480		Urban Tan 478	
	Min	Max	Min	Max	Min	Max
700	38	53	19	36	25	48
720	38	58	20	36	25	52
740	39	62	20	36	25	54
760	40	66	21	36	26	56
780	41	72	21	38	27	57
800	43	76	22	43	28	58
820	45	76	23	45	30	58
840	48	78	24	46	33	58
860	50	78	25	46	36	59

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TABLE II.G. Spectral reflectance (percent), Class 14.

Marine Pattern (MARPAT) Snow Camouflage Pattern 1/						
Wavelength, nanometers (nm)	White Snow 506		Light Snow Gray 507		Medium Snow Gray 508	
	Min	Max	Min	Max	Min	Max
600	80	80	46	66	28	36
620	80	80	47	66	30	40
640	78	80	48	66	30	44
660	78	80	49	68	34	44
680	78	80	50	72	36	48
700	78	80	51	72	40	56
720	78	80	52	72	40	56
740	78	80	53	72	42	56
760	78	80	54	72	44	56
780	78	80	55	74	46	58
800	78	80	56	74	46	60
820	78	80	57	76	48	64
840	76	99	58	76	48	66
860	76	99	59	76	50	66

1/ The Dark Snow Gray 509 color of the pattern is too small to measure spectral reflectance.

3.8 Physical requirements. The finished cloth shall conform to the requirements in Table III, when tested as specified in 4.5.3.

TABLE III. Physical requirements (all Types and Classes).

Characteristic	Requirements
Weight, oz./sq. yd.	
Minimum	3.8
Maximum	5.0
Yarns per inch, minimum	
Warp	80
Filling	56
Breaking strength (pounds), minimum	
Warp	275
Filling	225
Air permeability, ($ft.^3/min./ft^2$) (max)	
Type I	6.0
Type II	2.0

3.8.1 Weave. The cloth shall be constructed as a plain weave with one (1) up and one (1) down. Testing shall be as specified in 4.5.3.

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3.8.2 Width. For Government procurements only, unless otherwise specified, the width of the finished cloth shall be as specified in the contract or purchase order (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.9 Finish. Type II, Type III and Type IV cloths shall be given a water repellent treatment (see 3.9.1). Type IV cloth shall be back coated (see 3.9.2).

3.9.1 Water repellent (Type II, Type III and Type IV). The Type II, Type III and the face side of Type IV cloth shall be water repellent treated with a fluorocarbon treatment and shall conform to the requirements specified in Table IV. Testing shall be as specified in 4.5.3.

TABLE IV. Water repellency requirements (Type II, Type III and Type IV).

Characteristic	Requirements
Spray Rating Type II, Type III and Type IV Initial After three (3) launderings	1/ 100, 100, 90 80, 80, 70
Hydrostatic resistance, cm (min) Type II Initial After three (3) launderings	2/ 35 35
Resistance to organic liquids (min) Type II and Type IV Initial After three (3) launderings	3/ No wetting No wetting

1/ The results of the three (3) individual determinations on the sample unit for spray rating shall be equal to or better than the specified ratings when tested as specified in 4.5.3.
2/ No individual specimen shall fall below 30 centimeters.
3/ No wetting by n-tetradecane after 30 seconds initially and after three (3) launderings.

3.9.2 Coating treatment (Type IV). The Type IV cloth shall be coated on the back side only with a suitable clear polyurethane coating compound and shall show water repellency characteristics on the face side. If plasticizers are used in the coating, only phosphate or phthalate ester type plasticizers shall be used. The cloth shall conform the requirements in Table V when tested as specified in 4.5.3.

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TABLE V. Water repellency requirements (Type IV).

Characteristic	Requirements
Hydrostatic Resistance, psi minimum Initial After abrasion After strength of coating After high humidity	 60 35 35 1/ 40
Stiffness, cm (warp only) maximum: at 0°F	10
Adhesion of coating, lbs./2-inch width minimum	7 2/
Blocking, rating maximum	No. 1
Water vertical wicking, inch maximum	1/8
Resistance to leakage	No leakage
Resistance to Diethyltoluamide (DEET)	3/

1/ The cloth shall not become stiff and brittle nor soft and tacky and there shall be no evidence of cracking or crazing under visual examination.
2/ Applicable if a film or coating is applied to the surface of the cloth.
3/ The cloth shall show no lifting, no tackiness, no solution, no pickoff, no adherence to itself greater than scale rating two (2) (slight blocking).

3.10 pH. The pH value of the water extract of the finished cloth shall be no lower than 5.5 nor more than 9.0 when tested as specified in 4.5.3.

3.11 Dimensional stability. The finished cloth shall have an average dimensional change of no more than 3.0 percent in the warp and no more than 2.0 percent in the filling directions after one (1) laundering cycle. Testing shall be as specified in 4.5.3.

3.12 Seam efficiency. The finished cloth shall have a seam efficiency of no less than 85.0 percent when tested as specified in 4.5.3.

3.13 Toxicity. The finished fabric shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.14 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished full width in continuous lengths, each not less than 40-yards. Each length shall be put-up full width on a roll as specified in 5.1.

3.15 Fiber identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

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3.16 Workmanship. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the established maximum point value.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with (3.1), shall be inspected, examined for appearance, color and finished defects and tested for the characteristics as specified in Table VI.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.5 through 4.5.2 and the tests of 4.5.3 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4 and with acceptance quality limits as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.4 Inspections conditions. Unless otherwise specified, all inspections shall be performed in accordance with all the requirements of the referenced documents, unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents.

4.5 Yard by yard examination. Each roll in the sample shall be examined yard-by-yard on both sides. When the total yardage in the roll does not exceed 100-yards the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100-yards, only 100-yards shall be examined. All defects, as defined in ASTM D3990 and MIL-STD-1487 for Type IV cloth, that are clearly noticeable at normal inspection distance (3-feet) shall be scored and assigned demerit points as listed in 4.5.1 except that only those slubs and knots which exceed the limits shown on the Fabric Defect Replica Scale (See 2.3), “D” or “3-1/2” as applicable for slubs and “C” for knots, shall be scored and coarse yarn shall only be scored as a defect when the coarse yarn is twice the diameter of the normal yarn used in the fabric. No linear yard (increments of 1 square yard on the measuring device of the inspection machine) from any one (1) roll shall be penalized more than four (4) points. The sample size shall be selected at random, no less than 10 percent of the rolls per lot. The lot shall be unacceptable if the points per 100 square yards examined exceeds 20.0 points. The lot shall be unacceptable if the points per 100 square yards of two (2) or more individual rolls exceeds 30.0 points.

Point computation for lot quality and individual roll quality shall be as follows:

$$\text{Points per 100 square yards} = \frac{\text{Total points scored in sample x 3600}}{\text{Total yards inspected square yards}}$$

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4.5.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3-inches in any dimension - one (1) point
For defects exceeding 3-inches, but not exceeding 6-inches in any dimension - two (2) points
For defects exceeding 6-inches, but not exceeding 9-inches in any dimension - three (3) points
For defects exceeding 9-inches in any dimension - four (4) points

The following defects, when present, shall be scored four (4) points for each yard in which they occur:

Defect Description
Any hole, cut, pinhole, tear, scratch or abrasion mark 1/ Edge ravels when pulled outward
Uneven weave throughout
Pattern design not equal to the standard sample
Incorrect color in any part of the pattern
Pattern repeat not equal to the standard sample
Pattern repeat less than 33.50-inches or more than 37.25 in the warp direction for Class 9
Pattern repeat less than 22.75-inches or more than 26.50 in the warp direction for Class 10
Pattern width in filling direction more than 68-inches for Class 10
Pattern repeat less than 14.50-inches or more than 18.0 in the warp direction for Class 11
Pattern repeat less than 24.75-inches or more than 28.50 in the warp direction for Class 12 and 13
Pattern repeat less than 34.50-inches or more than 37.50 in the warp direction and less than 58.50-inches or more than 61.50 for the filling direction for Class 14
Skitteriness (mottled, uneven color) or pattern exceeds that shown by the standard sample
Excessive feathering or spew (fuzziness at color boundaries) of pattern as compared to the standard sample
Excessive haloing or trapping (overlaying of colors) of pattern as compared to the standard sample
Design not printed on face side of cloth
Any spot, stain or streak more than 1-inch in combined directions 2/ Overall application of water repellent not uniform (Type II, Type III and Type IV)
Uneven coating application, bubbling/blister, bare spot, discolored, brittle, tackiness (sticky to the touch) (Type IV)
Objectionable odor 3/ Baggy, ridgy, or wavy cloth
Any embedded foreign matter or overall uncleanness

1/ The same yardage shall be given a through-lighting inspection for pinholes and thinly coated areas.
2/ Clearly visible at normal inspection distance (3-feet) by either direct viewing or through lighting.
3/ Normal residual odor of chemicals used shall not be regarded objectionable.

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4.5.2 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. Any roll in the sample, off shade or shaded side to side, side to center, or end to end, or any roll that does not have the same appearance as the standard sample, shall be cause for rejection of the entire lot.

4.5.3 End item testing. The cloth shall be tested for the characteristics listed in Table VI. The methods of testing as specified wherever applicable and as listed in Table VI shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more sample units or the lot average for dimensional stability fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE VI. End item tests.

Characteristic	Requirement Paragraph	Test Method
Fiber Identification	3.4.1	AATCC 20 or ASTM D276 1/
Yarn	3.4.2	Visual
Visual shade matching	3.5.1	4.6.1
Colorfastness: Laundering (after three (3) cycles) Light (after 40 AFU or 170 kJ/(m^2nm) @420 nm) Accelerated Weathering (after 40 AFU or 170 kJ/(m^2nm) @340 nm) Crocking (Wet & Dry)	3.5.2 & Table I 3.5.2 & Table I 3.5.2 & Table I	4.6.2 AATCC-61, Test 1A 2/, 3/ AATCC-16.3 4/ AATCC-169, Option 4/ AATCC-8 5/
Spectral reflectance Classes 1 -6 Solid Shades Classes 9-14 Camouflage Pattern	 3.7 and Table II and Table II.A 3.7.1 and Table II.B - Table II.F	4.6.3
Weight Yarns per inch Breaking strength Air permeability (Type I and Type II)	3.8 & Table III 3.8 & Table III 3.8 & Table III 3.8 & Table III	ASTM D3776/D3776M, Option C ASTM D3775 ASTM D5034 ASTM D737

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TABLE VI. End item tests - Continued

Characteristic	Requirement Paragraph	Test Method
Weave	3.8.1	Visual
Spray rating (Type II, Type III and Type IV) Initial After three (3) launderings	3.9.1 & Table IV	AATCC-22 AATCC-135, Test (3)V Aiii, AATCC 22
Hydrostatic height: (Type II) Initial After three (3) launderings	3.9.1 & Table IV	AATCC-127 or ASTM D751 6/ AATCC-`135, Test (3)V, Aiii, AATCC-127 or ASTM D751 6/
Resistance to organic liquids (Type II and Type IV) Initial After three (3) launderings	3.9.1 & Table IV	AATCC-118 AATCC-135, Test (3)V, Aiii, AATCC-118
Hydrostatic Resistance, psi (min) (Type IV) Initial After abrasion After strength of coating After high humidity	 3.9.2 & Table V 3.9.2 & Table V 3.9.2 & Table V 3.9.2 & Table V	 ASTM D751 7/ ASTM D3886 8/ & ASTM D751 7/, ASTM D751 7/, 9/ ASTM D751 7/, 10/
Stiffness, cm (warp only) (max) (Type IV) at 0℉	3.9.2 & Table V	TAPPI T451 11/
Adhesion of coating, lbs/2 inch width (min) (Type IV)	3.9.2 & Table V	ASTM D751 9/, 12/
Blocking, rating (max) (Type IV)	3.9.2 & Table V	ASTM D751 13/
Water wicking, inch (max) (Type IV)	3.9.2 & Table V	14/
Resistance to leakage (Type IV)	3.9.2 & Table V	ASTM D751 15/
Resistance to diethyltoluamide (DEET) (Type IV)	3.92. & Table V	4.6.4
pH	3.10	AATCC-81
Dimensional stability	3.11	AATCC-135, Test (3)V, Aiii
Seam efficiency	3.12	ASTM D1683/D1838M 16/
Toxicity	3.13	4.7

1/ In case of dispute, the ASTM method prevails.
2/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
3/ Only the stain on the nylon and cotton fibers of the color transfer cloth shall be evaluated.
4/ The exposure cycle shall be half of the time exposed to light and half of the time exposed to water spray. Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change.
5/ Rated using the AATCC Evaluation Procedure 8, AATCC 9-step Chromatic Transference Scale

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6/ Procedure B, Procedure 2
7/ Hydrostatic resistance procedure A, procedure 1 with water pressure applied to the back side of the finished cloth.
8/ Except that a solid rubber diaphragm 0.030 (±0.01) inches thick shall be used. The abradant shall be faced side of the finished cloth. The face side of the finished cloth specimens shall be abraded 1000 multidirectional cycles. The specimen shall be tested for hydrostatic resistance with the abraded portion in the hydrostatic tester.
9/ Except a 20 pound shall be applied.
10/ Three (3) 4 by 4-inch specimen shall be laid flat, face side up, on a supporting plate and the assembly placed in a desiccator containing water in the lower portion. The water level shall be approximately 1-inch below the specimens. The lid of the desiccator shall be put in place and the desiccators placed in a circulating air oven having a temperature of 180° (± 2)°F for a period of 14 days. At the end of the aging period, each specimen shall be removed from the desiccator and tested immediately for hydrostatic resistance.
11/ Except that five (5) test specimen with long dimension parallel to the warp direction of the finished cloth shall be tested. The apparatus and test specimens shall be subjected to a temperature of 0°F (± 2)°F for a period of four (4) hours and the test performed in a still atmosphere at that temperature.
12/ Except that 2-inch wide reinforced coating adhesion specimens, cyanoacrylate (solventless) adhesive, and pulling clamp speed of 5 mm/sec shall be used. Three (3) specimens shall be tested.
13/ Except that only one (1) specimen shall be exposed at an oven temperature of 180°F (± 2) °F for 30 minutes.
14/ Three (3), 2-inch by 7-inch specimens (face side) with long dimension parallel to the filling direction of the finished cloth shall be tested. One (1) end of the each specimen shall be immersed to a depth of 2-inches in distilled water at room temperature for five (5) minutes. When withdrawn from the water, the height of the wetted area above the original 2-inch water line of each specimen shall be measured to the nearest 1/32-inch.
15/ Hydrostatic Resistance procedure B, procedure 2 with the hydrostatic head fixed at 50 centimeters and applied to the test specimen for 10 minutes. The back side of the finished cloth shall contact the water. The report shall only include the "measurement of the appearance of water drops". Leakage of any specimen shall be considered a test failure. Leakage is defined as the appearance of water at three (3) or more different places within the 4-1/2-inch diameter test area.
16/ The needle shall measure 0.040 (±0.001) inch across the blade at the eye. The thread for all
classes shall be polyester core, cotton- or polyester- covered in accordance with A-A-50199 Tex size 36-45, 2 ply for the needle and Tex size 31-35, 2 ply for the looper.

4.6 Tests and methods of inspection

4.6.1 Visual shade matching (all Classes). The color and appearance of the cloth shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500K (± 200) illumination of 100 (±20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

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4.6.2 Colorfastness evaluation testing. When testing for colorfastness properties, each color shall be evaluated, whenever possible, separately and reported as such. In cases where the print pattern does not allow for the evaluation of each color separately, the test results should indicated which colors were evaluated together.

4.6.3 Infrared spectral reflectance (measurements in the visible/near infrared). Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) for Universal Camouflage, OCP, MARPAT Woodland and most solids and 700 to 860 nanometers 3-color Desert and MARPAT Desert Camouflage at 20 nm intervals, as stated in the requirements, on a spectrophotometer (see 6.5) relative to a polytetrafluoroethylene (PTFE) family of compounds, the preferred white reference standard. Other white reference materials may be used, provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 26 nm at 860 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. The specimen shall be measured as a single layer backed with three (3) layers of the same fabric and shade for Woodland camouflage patterns and backed with four (4) layers for the Desert and Universal patterns. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6-inches away from the selvage. The specimen shall be viewed at an angle not greater than 10 degrees from normal with the specular component included. The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter for Classes 1 through 7 unless otherwise specified and 0.3725 inches in diameter or larger for Classes 8 through 12 unless otherwise specified (Always use the largest aperture possible). Photometric accuracy of the spectrophotometer shall be within 1 percent and wavelength accuracy within 2 nm. When the measured reflectance values for any color at four (4) or more wavelengths do not meet the limits specified in Tables II through Table IIF or as specified in the referenced documents it shall be a test failure.

4.6.4 Resistance to Diethyltoluamide (DEET). The DEET solution contains 75 percent diethyltoluamide and 25 percent ethanol (see 4.6.4.1). Three (3) drops of the DEET solution shall be placed in the center of a 4 by 8-inch specimen of the finished cloth with the coated side up. The specimen shall be folded to form a 4 by 4-inch square with the coated side folded onto itself. The folded specimen shall then be placed between two (2) 6 by 6-inch glass plates and a 4-pound weight placed on the assembly and left at standard conditions for 16 hours. The specimen shall then be removed from between the glass plates, scale rated for blocking as shown in ASTM D751, except that the specimen shall not be exposed to elevated temperature, and then immediately examined and rated against the blocking scale for conformance to the requirements in Table II.

4.6.4.1 DEET reagent. The DEET reagent is an insect repellent reagent solution of 75 percent by weight (min) of diethyltoluamide and the remainder denatured alcohol. The diethyltoluamide component of the solution shall be a technical grade and containing N, N-diethylmetatoluamide of not less than 95 percent purity and the remainder shall consist of entirely or mixture of ortho or para isomers of N, N- diethyltoluamide. The denatured alcohol component of the solution shall be ethanol, U.S.P. 94.9 percent by volume and denatured in

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accordance with The Code of Federal Regulations 27 CFR 21, Formula 40 (see 2.2.2). The diethyltoluamide should be registered with the U. S. Environmental Protection Agency (EPA) in accordance with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) rules and regulations for implementing Code of Federal Regulations 40 CFR 150 - 180 (see 2.2.2).

4.7 Toxicity test. Unless otherwise specified (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of the studies indicate the finished cloth is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.13) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This cloth is intended for use in the rucksack (light-weight), the jungle hammock, carrying case for the collapsible canteen, sleeping bag sack and parachute components.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Types and Classes of cloth required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2)
d. When a first article is required (see 3.1, 4.2, and 6.3).
e. Colors required (see 3.5).
f. Width of cloth required (see 3.7.2).
g. Length required if other than specified (see 3.15).
h. Toxicity requirements (see 3.13 and 4.7).
i. Conformance inspection (see 4.3)
j. Inspection conditions (see 4.4)
k. Packaging (see 5.1).

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6.3 First article. When a first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a pre-production sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Snow White 506 replaces Arctic White 488.

6.6 Fabric Defect Replica Scales are available for purchase from SDL Atlas, Contact Customer Service 1-803-329-2110 and request item number 402985 – Slub/Knot Replica Set.

6.7 Subject term (key word) listing.

Canteen, collapsible
Hammock, jungle
Parachute canopy
Parachute container
Rucksack
Sleeping bag stuff sack

Changes from previous issue. ASTM D747 (Standard Test Method for Apparent Bending Modulus of Plastics by Means of a Cantilever Beam) was removed from Section 2.3, Non-Government publications, due to it not being specified in any requirements, and FED-STD-4 (Glossary of Fabric Imperfections) was superseded by ASTM D3990 (Fabric Defects, Standard Terminology Relating to).

Custodians:
Army – GL
Navy – NU
DLA – CT

Preparing activity:
DLA-CT

Project 8305-2023-003

Review activities:
Army – MD
Navy – MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST online database at https://assist.dla.mil

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